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Selecting Wells for Underbalanced Coiled Tubing Drilling in Deep and Tight Reservoirs

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Abstract

Drilling conventional horizontal wells in deep, tight reservoirs using conventional methods poses several challenges. Recently, underbalanced coiled tubing drilling (UBCTD) has emerged as a promising alternative, particularly for smaller borehole diameters. UBCTD offers notable advantages, such as enhanced rates of penetration, reduced formation damage, and the potential to achieve natural production without stimulation. However, the practical criteria for determining the suitability of UBCTD for specific wells remain underexplored. To identify suitable wells for UBCTD, we have developed comprehensive workflow to provide foresight and guidance. The workflow focuses on determining whether a well should be drilled using UBCTD before the commencement of drilling operations. This workflow consists of several critical steps: (1) diagenetic rock typing; (2) natural fracture prediction; (3) 3D geomechanics modeling; (4) extracting rock properties and in-situ stresses along the planned well trajectory from the 3D geomechanics model; (5) breakdown pressure envelope calculation. If the calculated breakdown pressure is lower than the wellhead can safely provide and the rock quality is deemed inadequate for natural production, conventional drilling methods will be selected, followed by hydraulic fracturing treatments after drilling. Conversely, if the breakdown pressure is relatively high and diagenetic rock typing indicates favorable conditions for natural production, UBCTD may be recommended for drilling. The developed workflow and criteria have been applied retrospectively to analyze conventional wells that experienced challenges during hydraulic fracturing, as well as wells drilled using UBCTD. Field case studies will be presented to validate the effectiveness of the workflow and criteria.

Introduction

Drilling conventional horizontal wells in deep-seated and tight gas reservoirs can be very challenging, as it might face issues such as a slow rate of penetration and rapid abrasive wear on drilling bits. Generally, horizontal wells in deep and tight gas reservoirs are hydraulically fractured to improve well productivity. However, hydraulic fracturing can fail at the beginning if the perforations require high breakdown pressure to initiate fractures [1-4]. In such cases, underbalanced coiled tubing drilling (UBCTD) may be a viable alternative, provided the reservoir rock has good permeability. UBCTD allows for continuous drilling and

pumping, which can increase the rate of penetration and avoid pipe differential sticking. It minimizes formation damage and enhances production. For underbalanced drilling, the pressure in the wellbore is maintained at a level lower than the pore pressure of the formation being drilled. Generally, UBCTD wells do not plan to be hydraulically fractured after drilling. For this reason, underbalanced coiled tubing drilling can be an excellent choice for certain reservoirs, such as carbonate and sandstone formations, which often have natural fractures and relatively higher permeability compared to unconventional plays. In such cases, the well can produce naturally without requiring hydraulic fracturing or other stimulation methods after drilling.

In practice, the key issue is how to determine whether underbalanced coiled tubing drilling should be used for a well before drilling commences. To address this, the paper presents systems and methods for selecting underbalanced coiled tubing drilling, which consist of several components: 1D mechanical earth modeling (1DMEM), natural fracture prediction, 3D geomechanical modeling, calculation of breakdown pressure for perforation fracturing, classification of rock types, and recommendations for either UBCTD or conventional drilling with hydraulic fracturing. After UBCTD drilling, no further action is required if the well can produce naturally. In addition, the in-situ stress changes caused by drilling-induced rock fragmentation, a factor that has historically received limited attention in the oil and gas industry. These stress changes are then used to update the in-situ stress model and assess whether natural fractures along the wellbore will shear slip, potentially increasing the stimulated reservoir volume. Based on this analysis, combined with the rock's porosity and permeability, the suitability of UBCTD for a specific reservoir can be determined.

The developed workflow and criteria have been retrospectively applied to analyze conventional wells that faced challenges during hydraulic fracturing, as well as wells drilled using UBCTD. Field case studies will be provided to validate the effectiveness of these workflows and criteria. Overall, the proposed methodologies provide a logical explanation for phenomena observed in past operations and offer a reliable framework for selecting the most suitable drilling method or well type when planning new wells, particularly in deep and tight gas reservoirs.

An Integrated Workflow for Selecting UBCTD

The integrated workflow was developed and used to determine whether a well should be drilled using conventional drilling or underbalanced coiled tubing drilling (UBCTD) in deep and tight reservoirs. For conventional drilling wells, hydraulic fracturing is typically performed after drilling is completed. In contrast, UBCTD is specifically suited for high-quality rock with relatively high porosity, permeability, and diagenesis rock types, where no hydraulic fracturing treatment is initially planned. UBCTD wells are drilled with a smaller borehole diameter, with the goal of achieving natural production. If necessary, UBCTD wells can be hydraulically fractured using specialized technology. The key components of the workflow include: (1) Estimating geomechanical properties; (2) Image log processing for fracture intensity and maximum horizontal stress orientation; (3) 3D property modeling; (4) Natural fracture prediction (NFP); (5) Establishing a 3D geomechanics model that incorporates NFP for hydraulic fracturing modeling and fracture stability analysis; (6) Calculating the required breakdown pressure for clustered-perforation hydraulic fracturing treatment. If the required breakdown pressure is relatively low or the rock quality is poor, the well will be drilled conventionally with cemented casing, and hydraulically fractured later. However, if the required breakdown pressure exceeds the wellhead's capacity and the rock quality is relatively good, UBCTD is the preferred choice. For deep and tight gas reservoirs with high breakdown pressure, cased hole/perforation hydraulic fracturing treatment can be challenging to complete. In such cases, this workflow can help to determine whether UBCTD is a viable alternative.

Key Features of The Workflow

Formation Image Log Processing and Natural Fracture Prediction

Most reservoirs exhibit some degree of natural fracturing, regardless of whether they consist of shale, sandstone, or carbonate rock formations. Natural fractures should be a factor incorporated for determining the well types. Also for naturally fractured reservoir, hydraulic fracturing modeling should account for the interaction between hydraulic fractures and natural fractures. This needs to have a discrete fracture network (DFN) model embedded in the 3D geomechanics model for fracture stability analysis and hydraulic fracturing modeling. Figure 1 shows the step-by-step workflow for natural fracture prediction. Natural fracture prediction starts from wellbore formation image log processing and interpretation, which include the fracture orientations, dip angle, dip azimuth, and fracture types. The fracture point data will be used as input to predict the natural fracture distributions in the 3D space. The identification of natural fractures' point data holds significant importance in determining the necessity of conducting natural fractures prediction for a given field. Preceding the prediction, a geological model comprising horizons and faults needs to be established. Structural restoration follows, providing insights into the deformation between the current (deformed) and original (flattened top layer of each formation layer) configurations. Subsequently, strain and stress distributions are calculated utilizing paleo structure data such as faults and folds acquired from 3D structural restoration. The tectonic setting, characterized by strain/stress and orientation, acts as the fracture drivers throughout the grid, offering guidance for the 3D distribution of fracture intensity. These fracture drivers contribute essential information about the spatial extent of fractures, with fault-related fractures being predominant in subsurface observations, while other mechanisms like bending, stratigraphy mechanical contrast, compaction, and diagenesis play secondary roles. Following the characterization of fracture drivers, a fracture network model is constructed, employing either deterministic or stochastic approaches. Figure 2 shows an example model of predicted discrete fracture network, which contains various fracture sizes and orientations (azimuths, dips).

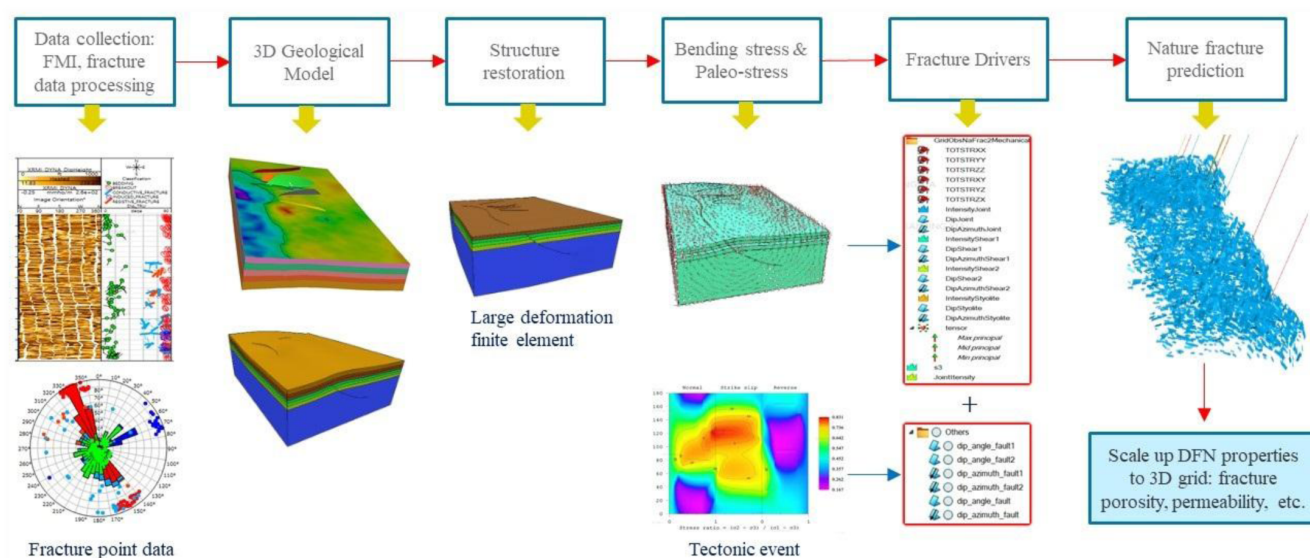


Figure 1—Workflow for Natural fracture prediction.

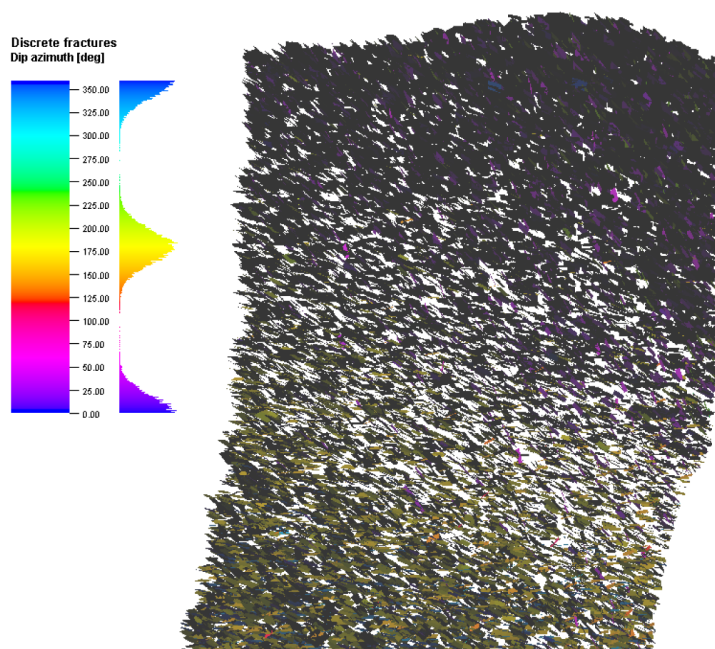


Figure 2—Discrete natural fracture network in the reservoir.

Rock Quality and Diagenetic Rock Types

The diagenetic rock typing is one of the main features of this workflow, which will be used to evaluate reservoir rock quality. This part is to generate different diagenetic rock types according to effective porosity and total clay content variation. Compaction and cementation are the two main diagenetic processes which reduce sandstone intergranular porosity. Compaction usually is a function of burial depth, and heavy compaction is expected with increased burial depth. Cementation is more important compared to compaction for the same sandstone reservoir since the heterogeneity was mainly caused by different types and amount of the cements. For example, a quartz-cemented clean sandstone has much better porosity and permeability than clay-rich sandstones, especially permeability in case of similar porosity. The clay content including illite, kaolinite and chlorite is critical for different diagenetic rock types. In this workflow, the total clay is used as one of the key factors for classifying sandstone diagenetic rock type.

The diagenetic rock typing criteria is based on sandstone petrophysical evaluation results, for example, the total clay content is less than 5% in clean sandstones. TYPE1 clean sandstones can naturally produce gas after drilling without fracturing (good for under balanced coil tubing drilling). Sandstone with more than 3% porosity and less than 10% total clay can contribute to production after hydraulic fracturing. Accounting for the above-mentioned factors, tight sandstone can be classified into six different types of diagenetic rock types (DRT), namely TYPE1 (code:6), TYPE2 (code:5), TYPE3 (code:4), TYPE4 (code:3), TYPE5 (code:2) and TYPE6 (code:1). Higher rock types codes have better rock quality, and will be better for production [5]. The study has shown that authigenic clay, especially authigenic illite, heavily affects the sandstone reservoir permeability, which will have a big impact on gas flow capacity. In order to estimate the gas flow capacity, a new flow index curve (FI) was proposed and created based on the petrophysical evaluation results. The FI can be expressed as:

$$FI = \frac{PHIE}{(a + TotalClay)}$$

Where a is a constant value, and total clay is the lump sum of illite, kaolinite and chlorite, and PHIE is the effective porosity.

Predicting In-Situ Stress Changes Due to Drilling Rock Fragmentation

In petroleum geomechanics, accurately estimating in-situ stresses is crucial for predicting the mud weight window and required breakdown pressure for hydraulic fracturing, among other applications. However, the changes in in-situ stresses around the wellbore due to drilling-induced rock fragmentation have been largely overlooked in the oil and gas industry. Typically, the far-field in-situ stresses are assumed to remain unchanged after drilling. To address this knowledge gap by simulating the wellbore stress changes induced by drilling rock fragmentation, which is essential for evaluating the potential for discrete natural fractures surrounding the wellbore to shear slip or reactivate. Since no analytical models are available for this purpose, a numerical method will be employed to simulate the in-situ stress changes. Figure 3 illustrates a numerical model designed to simulate the wellbore stress changes resulting from drilling rock fragmentation.

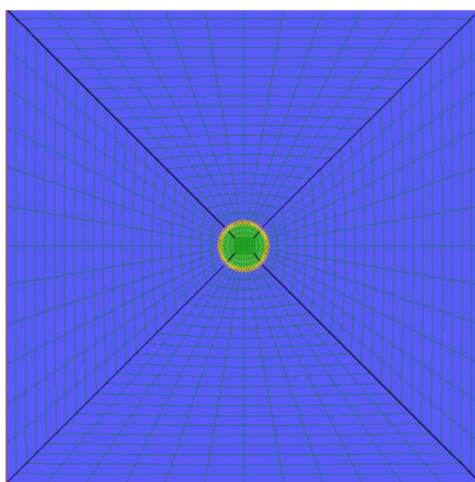


Figure 3—Finite element model for predicting stress changes due to drilling rock fragmentation.

The simulation process involves three distinct steps, as shown in Figure 4. The first step is to simulate the in-situ stresses. In the second step, the rock fragmentation process is simulated, assuming the rock remains in the borehole. At this stage, the rock is treated as a granular material without cohesion but with a frictional angle. The third step is to simulate the rock removal from the wellbore, a critical procedure that has been largely neglected in the oil and gas industry. This step is essential for accurately simulating the in-situ stress changes that occurs after drilling. Figure 5 presents the final in-situ stresses around the wellbore after the rock has been completely removed, which will be used to assess the stability of natural fractures. By accounting for the effects of drilling-induced rock fragmentation on in-situ stresses, which provide a more accurate and comprehensive understanding of wellbore behavior, enabling better decision-making in drilling and completion operations.

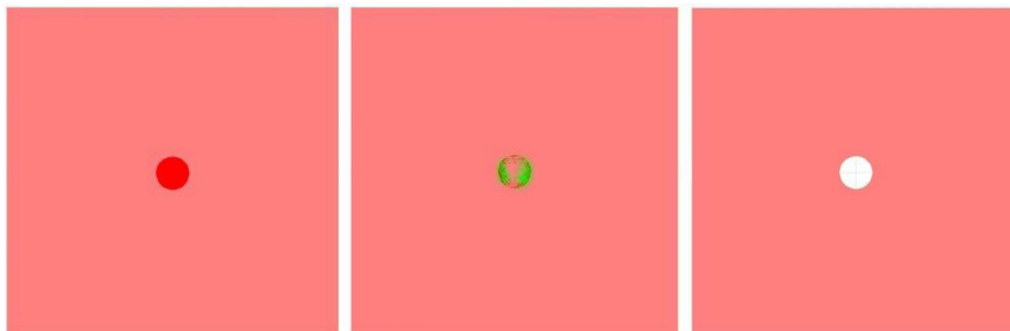


Figure 4—Three modeling stages for predicting stress changes due to drilling rock fragmentation.

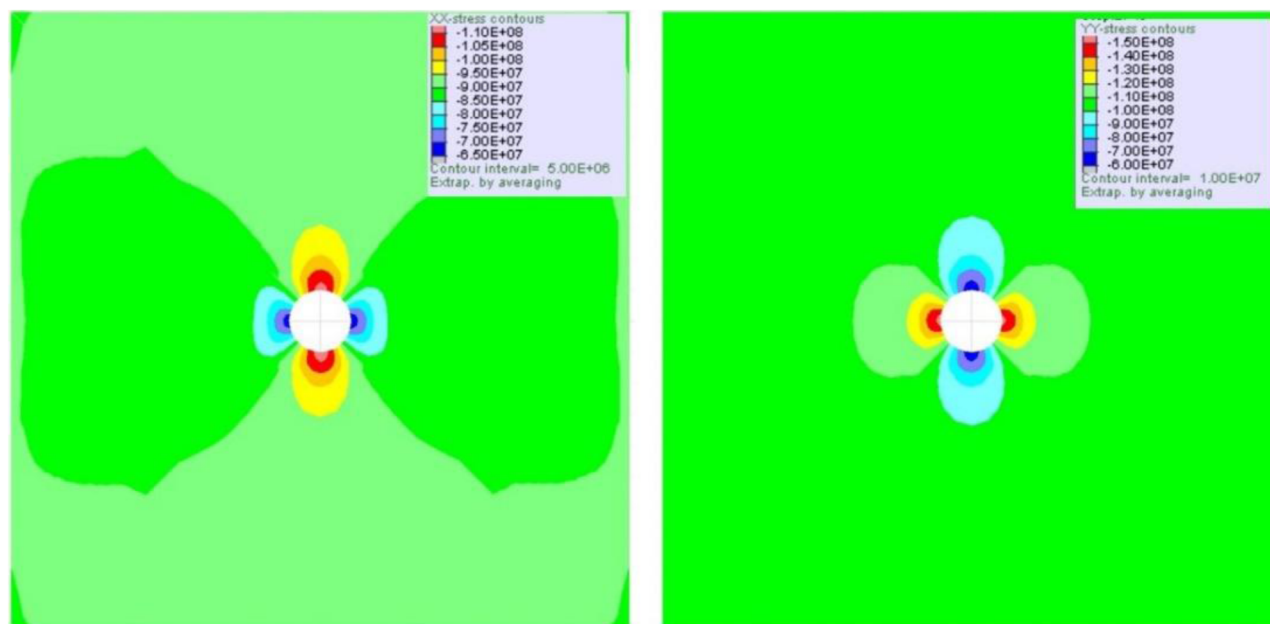


Figure 5—In-situ stresses after drilling rock fragmentation.

Evaluating Natural Fracture Stability Due to Drilling Rock Fragmentation

Natural fractures require changes in stresses and pore pressure to trigger shear slip, which occurs when the shear stresses acting on the fracture surfaces exceed the resistance to slip of the adjacent rock blocks. One of the primary mechanisms for triggering shear slip is the change in pore pressure caused by fluid injection [8]. The shear resistance is attributed to friction, which is proportional to the effective normal stress (i.e., the difference between the normal stress acting on the fracture and the fluid pressure) in the fracture. As long as the magnitude of the shear stress is lower than the shear resistance or frictional strength, the fracture remains in a stable state. Figure 6 illustrates the in-situ stress state around fractures and the impact of pore pressure changes on triggering fracture shear slip. The critical condition for shear slip to occur is governed by the Coulomb strength criterion, which embodies two fundamental concepts: friction and effective stress by

$$\tau = \mu(\sigma_n - p)$$

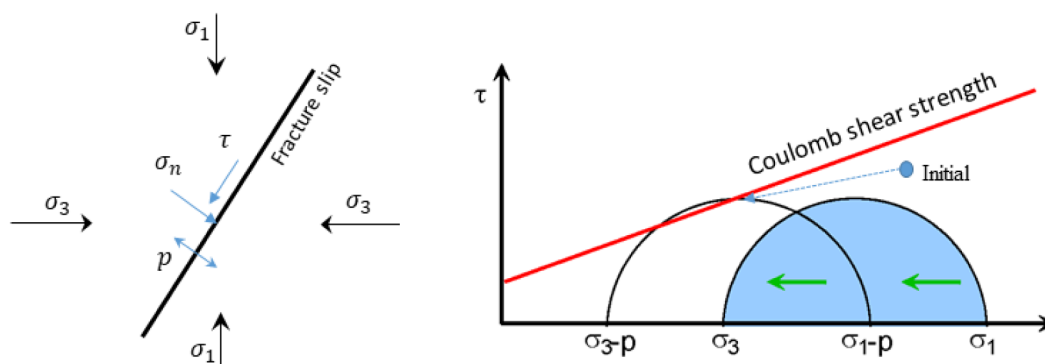


Figure 6—Coulomb stress change and shear strength.

The Coulomb criterion highlights the importance of effective stress, where the fluid pressure counteracts the effect of the normal compressive stress σ_n . This criterion reveals that fault slip can be triggered by one of three mechanisms: a decrease in normal stress, an increase in pore pressure, or an increase in shear stress. By considering these factors, the Coulomb stress change (ΔCSC) can be used to assess the stability of a natural fracture in response to changes in pore pressure and stress. The ΔCSC provides a quantitative measure of

the change in stress conditions that can either stabilize or destabilize a natural fracture. By evaluating the ΔCSC , we can predict whether a fracture is likely to become more stable or unstable due to changes in pore pressure and stress:

$$\Delta CSC = \Delta\tau - \mu(\Delta\sigma_n - \Delta p)$$

Where $\Delta\tau$ is the shear stress change on a fracture in the fracture direction (positive in the direction of fracture slip), $\Delta\sigma_n$ represents the compressive stress change that clamps or unclamps the fracture (positive if the fracture is in compression), Δp is the pore pressure change in the fracture that unclamps the fracture, and μ is the frictional coefficient of fracture surface. Based on the definition of ΔCSC , a positive change of ΔCSC promotes shearing slip and a negative change inhibits fracture shearing slip. Also the natural fracture stability can be evaluated using the following equation:

$$FVAL = \tau - [c + \tan\phi(\sigma_n - p)]$$

Figure 7 presents the results of a natural fracture stability analysis, which evaluates the impact of stress changes induced by drilling rock fragmentation on the stability of natural fractures. The analysis reveals that some natural fractures will undergo shear slip due to the stress changes caused by drilling rock fragmentation. This phenomenon is beneficial for enhancing permeability and increasing production rates after drilling, especially underbalanced coiled tubing drilling (UBCTD). The shear slip of natural fractures can create new flow paths and increase the connectivity of the fracture network, ultimately leading to improved reservoir performance [6,7,9]. By understanding the effects of drilling rock fragmentation on natural fracture stability, we can optimize their drilling and completion strategies.

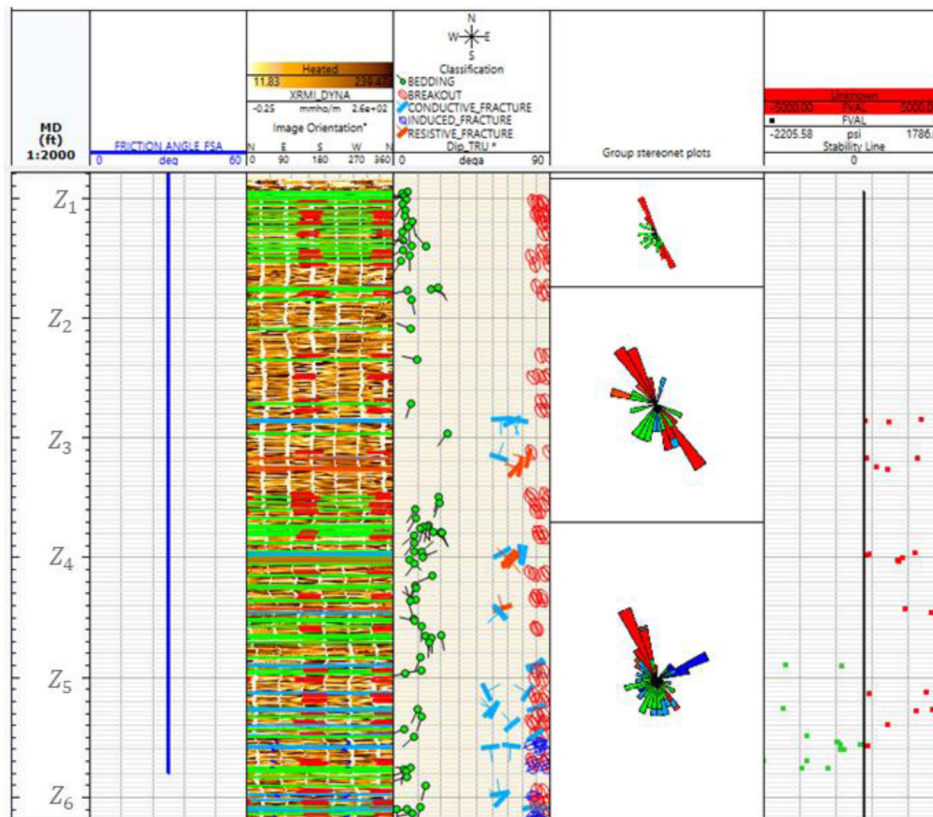


Figure 7—Natural fracture stability analysis due to stress changes induced by drilling rock fragmentation.

Field Example Study

Determining the right well type before drilling is crucial in practice. The well type and corresponding fracturing methods are closely linked, and selecting the optimal well type can significantly impact the success of the drilling operation. To achieve this, we can classify the rock type based on log data at well location, which enables us to evaluate the rock quality. Additionally, we can derive a 1D MEM model from a 3D geomechanics model, allowing us to calculate the required breakdown pressure as well as discrete natural fractures. By following our developed workflow, we can determine the most suitable well type. A retrospective study of three wells in the same field, as shown in Figure 8, demonstrates the effectiveness of our approach. The study includes three wells located in the same field and formation, with Well 1 drilled first, followed by Well 2. Notably, Well 1 has the best rock quality, and both Well 1 and Well 2 required hydraulic fracturing treatments to produce. The well 3 has relatively poor rock types compared to Well 2 and Well 3. This suggests that nearby wells in the same formation should be drilled using conventional drilling methods. In contrast, Underbalanced Coiled Tubing Drilling (UBCTD) is not a suitable well type in this case, as natural production is challenging to achieve after UBCTD drilling. The rock types, required breakdown pressure for Well 3 and post-fracturing production further validated the workflow prediction. Conventional drilling wells, which require hydraulic fracturing treatment, are a better option. Our field example studies validate the effectiveness of our developed workflow and criteria, providing a reliable approach for decision-making for placing new wells and determining the most suitable drilling method, particularly in deep and tight gas reservoirs.

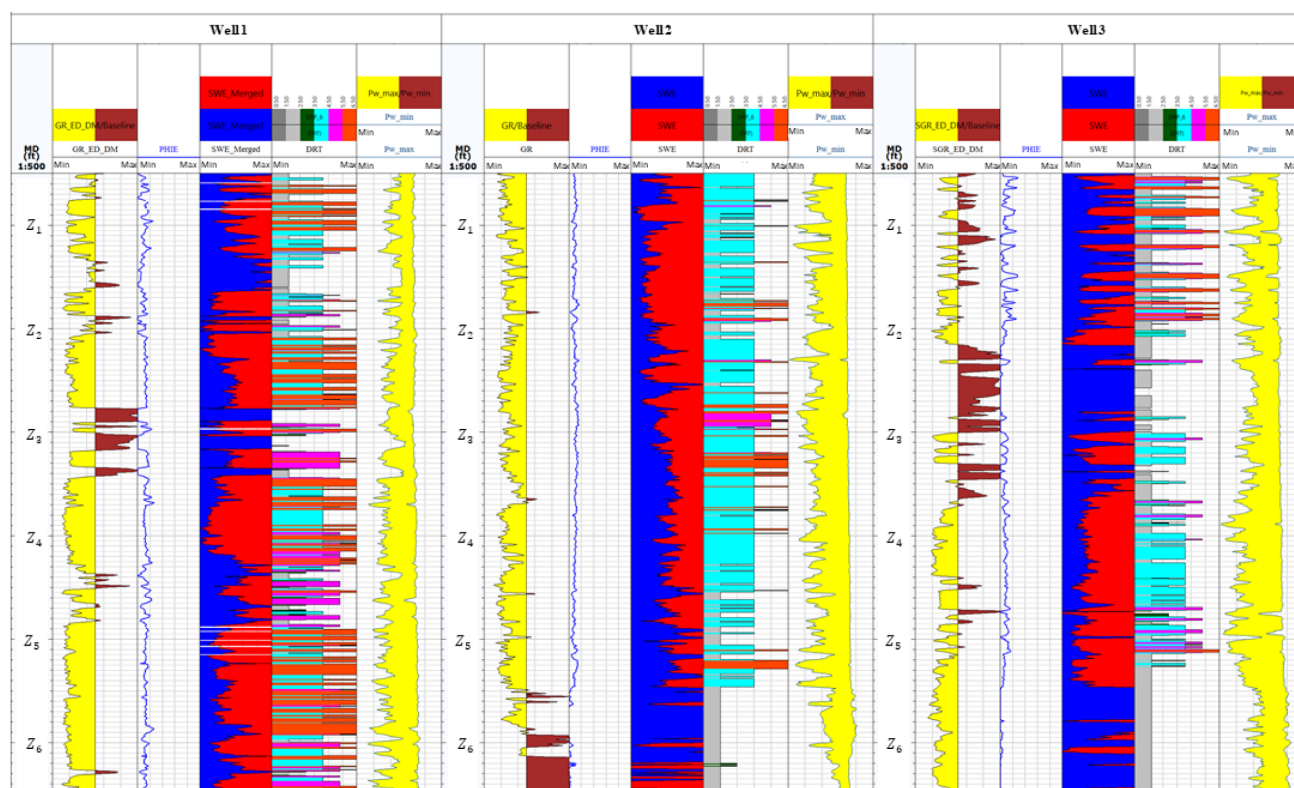


Figure 8—Determining well type based on the nearby wells' hydraulic fracturing and post-fracturing production performance.

Concluding Remarks

Deep and tight gas reservoirs used to heavily rely on hydraulic fracturing stimulation to achieve good production. However, conventional drilling wells stimulated by hydraulic fracturing treatment might not be best option. For reservoirs with favorable properties, such as good permeability, mobility, and numerous

natural fractures (as observed from image logs), underbalanced coiled tubing drilling (UBCTD) may be a viable alternative. To scientifically determine the optimal well type and corresponding stimulation methods, a comprehensive workflow is essential for guidance. To address this need, we developed a workflow for determining whether UBCTD is the right well type for a geologic setting. The workflow evaluates rock types, predicts natural fractures, and determines the required breakdown pressure for fracture initiation. This workflow helps determining whether a well should be drilled using UBCTD or conventional drilling. A field case study of three wells demonstrates the effectiveness of the developed workflow and criteria. By following the proposed workflow and procedures, field engineers can minimize uncertainties, saving time and money while increasing the chances of placing the right well types for good production and stimulating wells with optimal production rates.

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